

EDUCATION

Indian Institute of Technology Madras

B.Tech in Aerospace Engineering & Interdisciplinary M.Tech in AI and Data Science

('22 – '27)

CGPA: 9.14/10.00

PUBLICATIONS & CONFERENCES

- **Charukhesh Balaji Rakesh**, Ranganathan Srinivasan, R. Rengaswamy. "Endogenous Multi-Objective Optimization and Risk Assessment of Hybrid Renewable Energy Systems via GMM and Conditional Clustering" [Oral Presentation]. *International Conference on Renewable Energy Research and Applications (ICRERA) 2026, Paris, France*.
- **Charukhesh Balaji Rakesh**, Ranganathan Srinivasan, R. Rengaswamy. "Contractual and resource-aware capacity optimization for hybrid renewable energy systems (HRES) with storage" [Oral Presentation]. *GARI Research Symposium — International Conference on Renewable Energy 2026, Lisbon, Portugal*.
- **Charukhesh Balaji Rakesh**, Ranganathan Srinivasan, Prakash Dandoria, Atul Tomar, R. Rengaswamy. "Contractual and resource-aware capacity optimization for hybrid renewable energy systems (HRES) with storage." *Applied Energy (Elsevier)*. (Under review — Extended journal version of GARI 2026 paper)

RESEARCH & PROFESSIONAL EXPERIENCE

Fast & Precise Robotic Manipulation via Flow Matching and Latent World Models — [GitHub](#) (May '26 - Present)

Advisor: Prof. Aniket B | Purdue SURF Intern | IDEAS Lab | Purdue University

- Developing a generative trajectory planner via **Flow Matching** and **V-JEPA latent world model** for continuous 7D actions.
- Designing a **latent-space MPC** framework with closed-loop rollouts to mitigate drift in real-time **Robomimic** tasks.

Safety-Critical Trajectory Planning via Sequential Bayesian Inference and Stochastic Control (Aug '26 - Present)

Advisor: Prof. Raghunathan R | M.Tech Thesis | SENAI Lab | Indian Institute of Technology Madras

- Developing a real-time stochastic control framework integrating **CVaR**, **MPC** and **state estimation** for uncertain systems.
- Designing adaptive state-space models via **sequential Bayesian inference** to estimate **latent risk states** from forecasts.

LLM-Based Planner with 3D Scene Graphs for Long-Horizon Robot Manipulation — [GitHub](#) (May '25 - Dec '25)

Advisor: Prof. Guillaume A S | NUS IRIS Intern | MARMOT Lab | National University of Singapore

- Developed a long-horizon **LLM-based planner** augmenting **MobiPi**, using **3D scene graphs** to ground spatial reasoning.
- Translated **natural language instructions** into spatially consistent action sequences for complex tasks in **RoboCasa**.

Generative Temporal Modeling and Risk Assessment for Complex Physical Systems (Mar '25 - Dec '25)

Advisor: Prof. Raghunathan R | SENAI Lab | Indian Institute of Technology Madras

- Developed a **generative framework** via Δ -GMMs and clustering to synthesize statistically coherent **temporal sequences**.
- Quantified **downside risk** across synthetic environments via **Monte Carlo** rollouts bounded by **KL divergence** thresholds.

Risk-Aware Model Predictive Control under Sensor Uncertainty for UAV Navigation — [GitHub](#) (Aug '25 - Nov '25)

Advisor: Prof. Guruprasad K R | Vision & Intelligent Systems Lab | Indian Institute of Technology Kanpur

- Developed a **chance-constrained MPC** using **Gaussian Uncertainty models** to dynamically adapt safety margins.
- Achieved **12% faster paths** and **20% lower control effort** by replacing deterministic bounds with **probabilistic constraints**.

Adaptive Kalman Filtering with RLS-Inspired and Physics-Informed Methods — [GitHub](#) (Jan '25 - Apr '25)

Advisor: Prof. Srikrishna B | Communications Lab | Indian Institute of Technology Madras

- Developed a targeted injection **Kalman Filter** using χ^2 tests to dynamically adapt velocity covariance under disturbances.
- Integrated **RLS adaptation** with **physics-informed constraints** to achieve precise tracking and mitigate overshoot.

Data-Driven Discovery of Implicit Dynamics from Systems Under Feedback Control — [GitHub](#) (Jan '25 - Apr '25)

Advisor: Prof. Aniketh K | ACME Lab | Indian Institute of Technology Madras

- Developed an **implicit SINDy** and **RLS framework** via Two-Pass STLSQ to recover nonlinear physical dynamics.
- Decoupled the effects of plant friction from hidden **LQR gains** via **online RLS**, achieving **< 4% identification error**.

Contractual and Resource-Aware Capacity Optimization for HRES with Storage (May '24 - Mar '25)

Advisor: Prof. Raghunathan R | SENAI Lab | Indian Institute of Technology Madras

- Developed an **endogenous multi-objective solver** via **NSGA-II** to map design manifolds under non-linear constraints.
- Evaluated **pareto-optimal strategies** over a **15-year stochastic baseline** via **revenue-risk arbitrage** under uncertainty.

TECHNICAL PROJECTS

Thrust-Limited Sliding-Mode Guidance for Precision Soft Landing on Asteroids — [GitHub](#) (Apr '24 - Dec '24)

Course: Principles of Guidance for Autonomous Vehicles | Advisor: Prof. Satadal G

- Formulated a **sliding-mode guidance law** to execute precision soft landings under strict **non-convex thrust constraints**.
- Mitigated actuator instability via **dynamic thrust bounds** and control switching to achieve **fuel-efficient trajectories**.

Edge-Enabled Hardware-in-the-Loop (HIL) Control | EH Energy Group Systems Pvt Ltd (Nov '23 - Jan '24)

- Deployed **Hardware-in-the-Loop (HIL)** testbeds via **ESP32 edge-computing** to stream real-time physical telemetry.
- Implemented high-frequency control algorithms on **Beckhoff PLCs**, synchronizing hardware with **cloud monitoring**.

From-Scratch Transformer Architecture for Neural Machine Translation — [GitHub](#) (Jan '26 - Apr '26)

Course: Fundamentals of Deep Learning | Advisor: Prof. Ganapathy K

- Architected a **Transformer** with just **PyTorch** featuring multi-head attention, causal masking, and positional encodings.
- Engineered a custom **Noam learning scheduler** and analyzed attention-head specialization across semantic patterns.

Unified Multi-Task Vision for Fine-Grained Recognition & Segmentation — [GitHub](#) (Jan '26 - Apr '26)

Course: Fundamentals of Deep Learning | Advisor: Prof. Ganapathy K

- Developed a **unified PyTorch pipeline** using a shared **VGG11 encoder** and task-specific heads for 37-class prediction.
- Designed a **U-Net decoder** with skip connections, optimizing **IoU** and **Dice losses** for robust pixel-wise segmentation.

LEADERSHIP & FOUNDING EXPERIENCE

Co-Founder & Computer Vision Lead (Apr '24 - Present)

Student Unmanned Aerial Systems — IITM (Aerial Robotics) | [Website](#)

- Co-founded a UAV team, engineering a **quadcopter** with a full **ROS** stack for autonomous mapping and payload delivery.
- Led computer vision, training **YOLOv11** and **SAHI** pipelines on synthetic **AirSim** data for real-time small object detection.

Core Engineer & Control Lead (May '23 - Mar '24)

Pravahan — IITM (Marine Robotics with Hybrid Powertrain & Autonomous Hydrofoil)

- Designed a **Simulink** hybrid **fuel-cell & battery powertrain** featuring real-time load control to optimize energy efficiency.
- Developed an **autonomous hydrofoil controller** via **PID gain tuning** to regulate angle of attack for desired lift generation.

TEACHING, COURSEWORK & TECHNICAL SKILLS

Teaching Assistantships: AI in Predictive Maintenance; Mathematical Foundations of Data Science;

Key Coursework:

Robotics & Control: Principles of Guidance for Autonomous Vehicles; Robot Control; Data Driven Modeling of Aerospace Systems and Complex Fluid Flows; Control & Estimation for Engineering Systems*;

AI & Robotics: Foundations of Machine Learning; Introduction to Deep Learning; Data Analytics Lab; Machine Learning Operations (MLOps); Data Science & AI in Finance;

Math & Compute: Linear Algebra; Complex Variables; Differential Equations; Mathematical Foundations of Data Science; Introduction to Scientific Computing; Signals & Systems; Adaptive Signal Processing; Foundations of CFD;

Programming & Skills:

Languages & Hardware: Python, C++, MATLAB, Simulink, Bash, SQL, Octave, $\text{\LaTeX}$, ESP32, Beckhoff PLCs

Frameworks & Simulators: PyTorch, Tensorflow, OpenCV, ROS 1/2, MuJoCo, PyBullet, AirSim, RoboCasa, Robomimic

HONORS & ACHIEVEMENTS

- **Field Deployment, Indian Army (J&K Rifles):** Deployed real-time drone-based computer vision for reconnaissance.
- **Global Research Fellowships:** Selected for Purdue SURF (**Top 3%**) and NUS IRIS (**Top 1%**) international cohorts.
- **1st (2026) & 8th (2025) Place Globally, SUAS:** Ranked out of 70+ international teams in autonomous UAV technical design.
- **National 2nd Runner-Up, James Dyson Award (2024):** Engineered a kinetically-charged 5-stage UV filtration prototype.
- **Caterpillar Tech Challenge (2025):** Built an ML-based PID tuning system, placing top 2% of 1,500+ teams in India.

EXTRACURRICULAR ACTIVITIES

- **Student Mentorship:** Mentored first-year undergraduates at IIT Madras to facilitate their academic and campus transition.
- **Strategy & Puzzles:** Active FIDE-rated tournament chess player and World Cube Association (WCA) speedcuber.
- **Athletics (Badminton):** Gold Medalist at the '25 - '26 IIT Madras Schroeter Inter-Hostel Sports Championship.
- **Community Outreach:** Volunteered at Green Hope Foundation to provide tutoring and mentorship to orphaned children.